Activity Selection

Topic: Greedy File: g1.pdf

PROBLEM

Given activities with start and finish times, select the largest number of mutually non-overlapping activities.

GREEDY CHOICE

Sort activities by finish time. Select the activity that finishes first, then select each next activity whose start time is at least the last finish time.

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sort activities by finish time
choose the first activity
for each remaining activity:
    if start >= finish_of_last_chosen: choose it
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EXAMPLE

For intervals (1,2), (2,4), (3,5), (5,7), a valid maximum selection is (1,2), (2,4), (5,7).

COMPLEXITY

Sorting takes O(n log n); the scan takes O(n). Overall time: O(n log n).

WHY IT WORKS

Choosing the earliest finishing compatible activity leaves as much remaining time as possible for later activities.